Solution: Top 'K' Frequent Numbers

Problem Statement

Given an unsorted array of numbers, find the top 'K' frequently occurring numbers in it.

Example 1:

```
Input: [1, 3, 5, 12, 11, 12, 11], K = 2
Output: [12, 11]
Explanation: Both '11' and '12' apeared twice.
```

Example 2:

```
Input: [5, 12, 11, 3, 11], K = 2
Output: [11, 5] or [11, 12] or [11, 3]
Explanation: Only '11' appeared twice, all other numbers appeared once.
```

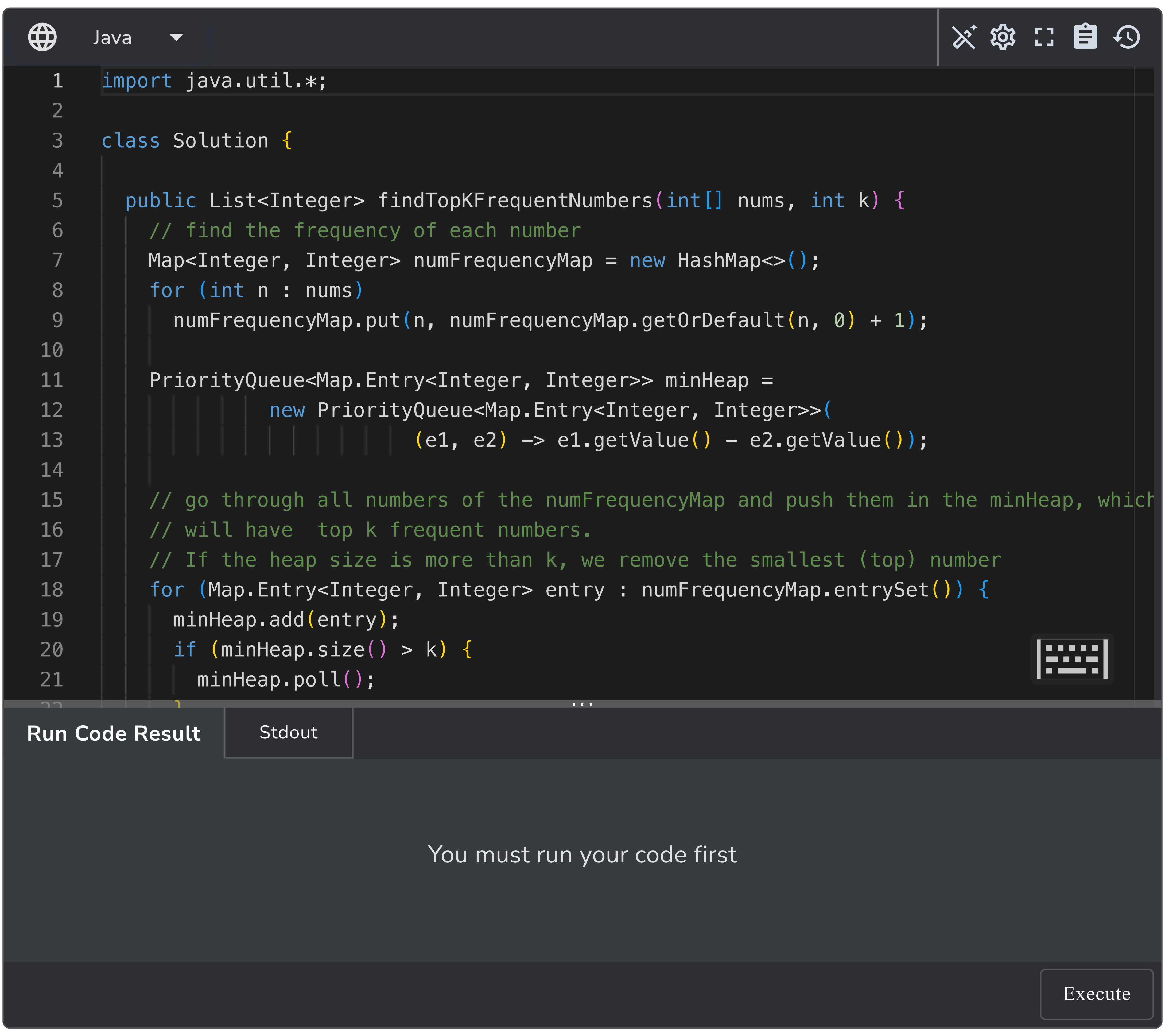
Solution

This problem follows `Top 'K' Numbers`. The only difference is that in this problem, we need to find the most frequently occurring number compared to finding the largest numbers.

We can follow the same approach as discussed in the **Top K Elements** problem. However, in this problem, we first need to know the frequency of each number, for which we can use a **HashMap**. Once we have the frequency map, we can use a **Min Heap** to find the 'K' most frequently occurring number. In the **Min Heap**, instead of comparing numbers we will compare their frequencies in order to get frequently occurring numbers

Code

Here is what our algorithm will look like:



```
import java.util.*;

class Solution {

  public List<Integer> findTopKFrequentNumbers(int[] nums, int k) {
    // find the frequency of each number
    Map<Integer, Integer> numFrequencyMap = new HashMap<>();
    for (int n : nums)
      numFrequencyMap.put(n, numFrequencyMap.getOrDefault(n, 0) + 1);

    PriorityQueue<Map.Entry<Integer, Integer>> minHeap =
              new PriorityQueue<Map.Entry<Integer, Integer>>(
                          (e1, e2) -> e1.getValue() - e2.getValue());

    // go through all numbers of the numFrequencyMap and push them in the minHeap, which
    // will have  top k frequent numbers.
    // If the heap size is more than k, we remove the smallest (top) number
    for (Map.Entry<Integer, Integer> entry : numFrequencyMap.entrySet()) {
      minHeap.add(entry);
      if (minHeap.size() > k) {
        minHeap.poll();
```

You must run your code first

Time Complexity

The time complexity of the above algorithm is $O(N + N * logK)$.

Space Complexity

The space complexity will be O(N). Even though we are storing only 'K' numbers in the heap. For the frequency map, however, we need to store all the 'N' numbers.

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